Jacob Rosario

Software Engineer

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Relevant Experience

Software & Embedded Systems Developer (Contract) **Oct 2026 - Present**
Canadian Athlete Development *Vancouver, BC*

- Developing C++ firmware for custom IMU hardware and app prototypes for jump frequency, estimated height and CSV export.
- Evaluating jump detection with Python and C++ replay tools across 4,800 simulated jumps and 9,000 non-jump scenarios, including regression tests for sensor faults and session controls.
- Exploring lower-cost volleyball jump tracking by adapting existing consumer wearables with custom motion-processing software and detection algorithms.

Software Developer & Sports Analyst Volleyball Stat Tracker **Feb 2026 - May 2026**
Hustle Volleyball Club *Burnaby & Vancouver, BC*

- Deployed a React app used courtside on iPad across 20+ competitive matches, replacing handwritten tracking with live statistics, visual summaries, and a coaching whiteboard.
- Refactored redundant application state into a single source of truth, eliminating state drift and improving data consistency.
- Worked directly with coaches to translate match-day needs into software features and use performance data to inform lineups, practices, and in-game adjustments; contributing to team's qualification for nationals.

Projects

RainCouver (Rain-Cover Navigation) | React, TypeScript, LiDAR/OSM data RainCouver **Sep 2026 - Present**

- Building a PWA that routes pedestrians through Downtown Vancouver by maximizing covered walking paths, using LiDAR-derived cover data across 25,000+ street segments and 3,000+ searchable places.
- Designing a self-improving routing system where user corrections update cover data in real time without disrupting an active route, addressing GPS drift and ambiguous-turn edge cases along the way.

RepPower | Swift, SwiftUI RepPower **Sep 2026**

- Developed an iOS application that uses AirPods motion-sensor data to estimate barbell squat velocity for velocity-based training.
- Engineered a Swift physics pipeline that transforms raw motion data into vertical acceleration and velocity while accounting for sensor noise and integration drift.
- Implemented zero-velocity drift correction and optimized high-frequency processing to maintain stable real-time measurements without blocking the SwiftUI interface.

Spike Timer | JavaScript, React Spike-Timer **Mar 2026**

- Built a client-side video-analysis tool that measures spike timing from user-marked takeoff, contact, and landing frames.
- Selected a lightweight geometric approach instead of pose-detection ML, enabling responsive browser-based processing.

Other Experience

Performance Analyst & Assistant Coach **Sep 2025 - Dec 2025**
Vancouver College *Vancouver, BC*

- Analyzed 5+ matches/week using video review and statistical data to identify opponent tendencies and performance patterns.
- Translated video data into actionable recommendations for lineups, opponent-specific strategies, and player development.

Volleyball Content Creator | @heartoverheight.vb **May 2025 - Apr 2026**

- Earned over 1 million views on multiple volleyball posts showing that shorter athletes can succeed as spikers.
- Secured sponsorships with Way of Wade and VKTRY Insoles; brands with a reported social audience of 1.6 million.

Barista **Sep 2023 - Sep 2025**
Starbucks *San Francisco, CA & Vancouver, BC*

- Processed 100+ transactions per shift at high-volume locations while maintaining accuracy and service quality under pressure.

Skills

- **Languages**: JavaScript, TypeScript, Python, Swift, HTML, CSS, C++ (prototyping)
- **Frameworks & Libraries**: React, SwiftUI, Three.js
- **Applied Engineering**: Geospatial Routing, Offline PWAs, Motion-Sensor Processing, Embedded Systems & Prototyping
- **Tools & AI**: Claude, Codex, Git, GitHub

Education

University of British Columbia **Jun 2025**
Bachelor of Science (BSc), Biology *Vancouver, BC*

- **Achievements:** Dean's List

Hobbies & Interests

Competitive volleyball, coaching & performance analysis | Sports analytics & biomechanics | AR/VR & 3D technology | Video games | Weightlifting & Olympic lifting | Content creation & video editing | Anime & manga | Playing guitar